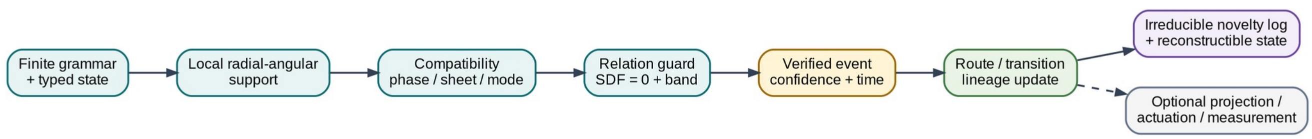
UNIFIED GEOMETRIC-TOPOLOGICAL SUBSTRATE

GPU-native compilation, physical-device architecture, performance, memory and compression addendum

UGTS-GN 1.1



Canonical query-first substrate: support, compatibility, guard, event, transition and lineage.

> **DELIVERABLE SCOPE**
>
> 197 normalized mechanisms from 10 source records; four compiled SPIR-V compute modules; direct Vulkan execution; 18 passing reference tests; GPU and physical hardware targets without Unity or Godot adapters.

Date: 15 August 2026

Evidence boundary: source-derived operators are separated from engineering translations, benchmarks and rejected claims.

Contents

READING RULE

The core is not a renderer. It is a directly queryable state-and-event substrate. Projection, display, print and actuation are downstream endpoints. [S03 pp. 5-12]

1. Executive synthesis

This addendum integrates the newly supplied glyph-to-bit document into the earlier unified corpus and changes the implementation target. The authoritative computational path is now direct Vulkan/SPIR-V compute or a measured physical device. Game-engine bridges are intentionally absent from the package.

The clean technical substrate is a finite grammar over typed state. Local radial-angular support removes irrelevant candidates; compatibility determines whether co-located states can couple; a relation or guard defines a transition surface; the earliest valid crossing becomes an event; transition logic updates route, topology and lineage; only irreducible novelty must be logged. This is the mature architecture stated by the corpus itself. [S03 pp. 5-12]

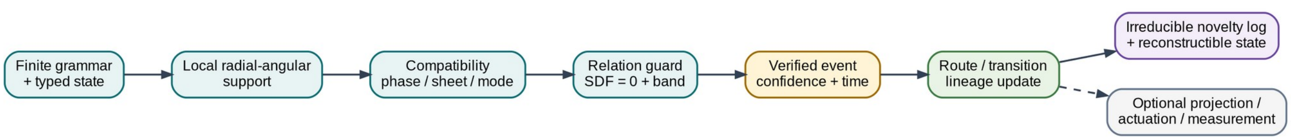


Figure 1. Canonical unification. Rendering and actuation are optional consumers, not the state authority.

1.1 What was added

Addition	Technical role	Disposition
Versioned glyph encoder	Separates ordinary 100%=1 from the source-defined 100100_2=36 transform.	Bounded, schema-controlled
Two-pulse timeline	Treats bit positions 5 and 2 as two address pulses separated by three positions.	Retained as encoding/address structure
Log-polar/SDF/BST register model	Provides coordinate addressing, guard evaluation, one-bit route/jitter metadata and branch selection.	Translated into typed GPU fields
Trident/Pythagorean cup	Three channels converge at a threshold and trigger a reset/drain.	Retained as hysteretic state-machine pattern
GPU-native target	GLSL 450 -> SPIR-V 1.4 -> direct Vulkan compute pipeline.	Implemented and validated on named software Vulkan device
Physical-device target	Fixed-function support/compatibility/guard/commit pipeline plus optofluidic endpoint.	Architecture specified; lab validation required

1.2 Concrete deliverables

- 197 normalized mechanisms with source basis, technical definition, GPU realization, physical realization, disposition and validation state.
- Four compiled SPIR-V modules: G64/E32 evaluate, G64/E32 evaluate+commit, G32/E16 evaluate and G32/E16 evaluate+commit.
- A direct Vulkan runtime with device timestamps, cold and cache-seeded pipeline creation, pipeline-cache reload and CPU-oracle validation.
- A dependency-free Python oracle, binary record packers, JSON schema, examples and 18 automated tests.
- A fixed-function Q16.16 RTL reference gate, register map, measurement plan and optofluidic architecture.
- Raw benchmark results, named metrics, memory statistics, compression models and plots.

BENCHMARK BOUNDARY

The validation device is SwiftShader Device (Subzero) (Vulkan 1.3.0, device type CPU/software). The run proves API-native execution and measured performance on that named device. It does not prove physical-GPU, FPGA, ASIC or photonic speed.

2. Evidence and normalization discipline

The source corpus contains exact arithmetic, useful construction rules, visual analogies, speculative physical claims and self-corrections. A clean substrate cannot silently treat all of them as equally established. Every extracted item therefore receives a disposition: RETAIN, TRANSLATE, CORRECTED, BOUNDED, OPTIONAL, DEMOTE or REJECT.

2.1 Admission rule

A motif enters the technical core only when it can be represented as a typed field, coordinate transform, support predicate, relation surface, compatibility predicate, transition/routing map, lineage rule, calibrated transfer function or measurable resource/error quantity.

Disposition	Meaning	Count
RETAIN	Kept as a direct operator or exact construction.	129
TRANSLATE	Converted from corpus language into a typed/measurable operator.	37
CORRECTED	Preserved after an explicit arithmetic, dimensional or terminology correction.	10
REJECT	Excluded from requirements because unsupported or absolute.	9
BOUNDED	Valid only under a declared schema, relation family or error range.	7
OPTIONAL	Useful downstream projection or implementation path.	3
DEMOTE	Retained as analogy or research prompt, not core mechanism.	2

2.2 Source-derived versus engineering-derived

Layer	Contains	Examples
Source-derived	Operators and structures explicitly present in the PDFs.	Radix thresholds, Pascal parity, cut-and-unroll glyph morphs, parity hinge, SDF boundary, double-vacuum non-coupling, hourglass routing, waveguide B.C.E.
Engineering normalization	A precise implementation that preserves the source motif while adding types and bounds.	G64/E32 and G32/E16 ABI, Vulkan descriptors, Q16.16 squared guard, event FIFO, schema versioning.

Layer	Contains	Examples
Measured package evidence	Results produced by the bundled implementation.	SPIR-V sizes, pipeline creation, dispatch latency, candidate/event rates, memory and validation flags.
Rejected/unsupported	Claims that the sources do not technically establish.	Zero latency/heat/memory, one bit as total state, physical Klein self-assembly, 202.74 J from 0.680 bits, universal O(1).

2.3 Privacy and source handling

The package does not redistribute raw extracted text from the uploaded PDFs. It includes hashes, privacy-safe source notes, normalized mechanisms and a source register. Personal identifiers and unrelated narrative/political passages were excluded from the synthesis.

3. Canonical substrate architecture

The mature corpus defines the substrate as a relation algebra and event calculus. The geometric vocabulary supplies local supports, phase sheets, boundaries and routing. The information architecture supplies typed state, compatibility, lineage and novelty. The implementation supplies direct query and measured event execution.

3.1 State model

```
q = (position, time, phase, sheet, orientation,
     generative_address, branch, policy, uncertainty)
```

The bundled hot-state profiles implement the subset needed for radial-angular support, topology/mode compatibility, a spherical relation guard, confidence filtering and lineage hashing. The full generative address and irreducible event log remain outside the compact hot record.

3.2 Query kernel

```
r = length(position)
cos_to_axis = dot(position/r, normalize(axis))
in_support = (r <= radius) and (cos_to_axis >= cone_cos)
compatible = mode_bit and sheet match and orientation match
sdf = r - radius
guard = abs(sdf) - epsilon
confidence = 2^(-32*abs(sdf))
verified = in_support and compatible and guard<=0 and confidence>=floor
```

The confidence expression is a deterministic benchmark law. It is not claimed as a universal probability model. Physical devices replace it with calibrated confidence or signal-to-noise semantics.

3.3 Topological operators

Corpus structure	Normalized operator	Technical consequence
Double vacuum	Co-located but incompatible phase/sheet/address sectors.	Position equality does not imply coupling.
Möbius/Klein motif	Explicit quotient/gluing map with orientation reversal.	Topology is a routing rule, not material magic.
Four-chamber hourglass	Four-way routing partition with a central event locus.	Transition table selects a chamber/branch at a guard crossing.
Parity hinge	One-bit route/template selector between declared topologies.	Bit has a narrow role; continuous state and lineage remain separate.
SDF = 0	Event surface with a finite guard band and numerical policy.	Boundary crossing invokes event semantics rather than endless sampling.
Cone/sphere/shell	Local support and domain-of-dependence operator.	Analytic pruning precedes relation solving.

3.4 Identity and persistence

Coordinates are not identity. Persistent identity uses a generative address, invariant history and ordered novelty events. Closed dynamics may be recomputed from seed and grammar; exogenous events must be stored. This is the basis for the package memory models and the distinction between dense state, compact novelty and event-log-only retention.

4. Addendum: glyph, timeline and threshold structures

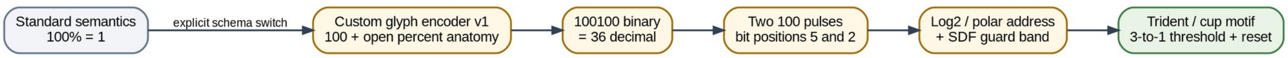


Figure 2. The new source is admitted as an explicit encoder and threshold-control family, not as ordinary arithmetic or free-energy physics.

4.1 Standard and custom semantics

NON-NEGOTIABLE DISTINCTION

In ordinary mathematics, 100% = 1. The value 36 appears only after selecting a custom source-defined glyph encoder. The schema records the encoder variant so the transform can never masquerade as ordinary percentage arithmetic. [S01 pp. 7-9]

Encoder	Bit sequence	Decimal	Status
standard	1	1	Ordinary arithmetic
glyph-100101-v0	100101	37	Earlier source variant
glyph-100100-v1	100100	36	Canonical addendum variant

```
glyph-100100-v1:
(1 << 5) | (1 << 2) = 32 + 4 = 36
```

4.2 Two-pulse address/timeline

The sequence 100100 contains two identical 100 groups. The addendum treats them as two pulses at bit positions 5 and 2. Technically, this is a sparse bitset or two-event address pattern. The separation is three bit positions; it does not imply physical time unless a clock and unit are declared. [S01 pp. 8-10]

4.3 Pythagorean and angular geometry

The source uses the scaled 5-12-13 triangle (1.5, 3.6, 3.9). The leg ratio is 3.6/1.5 = 2.4, giving an angle atan(2.4) approximately 67.38 degrees. The package retains this as a parameterized cone-angle example, not a universal constant.

4.4 Fractional shifts and de-phiming correction

Multiplication by 1.5 is not an integer left shift. In a base-2 logarithmic coordinate it is an additive displacement log2(1.5) = 0.5849625007. The source value 0.680 bits -> 202.74 J is dimensionally unsupported and is rejected. An engineering conversion from erased bits to a minimum heat requires a physical temperature and process; even then the Landauer result is a lower bound, not actual device energy.

4.5 Trident and Pythagorean cup

The trident/pinion and Pythagorean-cup material is technically useful when normalized as a three-to-one threshold/reset system: three channels converge, an accumulator rises, a crest crossing latches an event, the state resets or drains, and hysteresis prevents chatter. [S01 pp. 27-29]

```
IDLE -> ACCUMULATE -> THRESHOLD_CROSS -> EVENT_LATCH
     -> RESET/DRAIN -> HYSTERESIS_CLEAR -> IDLE
```

PHYSICAL BOUNDARY

A Pythagorean cup is a siphon driven by gravity and pressure. It does not create an absolute vacuum, destroy information, recycle energy without loss or supply a self-sustaining topological loop.

5. Unified mechanism taxonomy

The complete catalog contains 197 atomic mechanisms. The entries are deliberately small: each mechanism is one reusable numeric, geometric, topological, event, memory, GPU or hardware operator. The appendix gives the complete source-by-source traceability.

Domain	Count	Representative mechanisms
GPU-native	23	GLSL-to-SPIR-V compilation, Direct Vulkan compute runtime, Descriptor ABI, 256-thread workgroup, G64 authoritative state record, E32 full ev...
Physical hardware	23	Liquid overclad, Liquid-core guide, Input liquid lens, Solid photonic substrate, 2×2 interferometric kernel, Balanced detection, Spherical-t...
Topology	16	Möbius orientation reversal, Klein gluing map, Same coordinate, different sheet, Non-orientable orientation bit, Projected self-intersection...
Audit	14	100% does not ordinarily equal 36, 100101 versus 100100 inconsistency, Information entropy is not automatically heat, Absolute single vacuum...
Numeric/encoding	13	Radix digit threshold, Positional power weighting, Literal binary glyph parse, Left-shift pulse placement, Zero-based offset/ordinal distinc...
Projection	13	Cartesian-to-log-polar mapping, One-bit sector activation, Phasor coverage accumulation, Subpixel gradient alignment, Pulse-density modulati...
Kinematics	10	Stem as cutting/rotation axis, Side-view pyramid as delta triangle, Delta-T changing axis, Loop-to-line release, Localized parity hinge, Spl...
Event calculus	9	Relation surface, Support predicate, Compatibility predicate, Earliest valid event, Transition operator, Finite guard band, Confidence thres...
Glyph encoder	7	Standard percent semantics boundary, Percent loop decomposition, Slash/phi removal operator, Open-loop terminal suppression, 100100 concaten...
Field algebra	6	SDF zero as event boundary, CSG union, CSG intersection, CSG subtraction, SDF gradient orientation, Parity-conditioned field blend...
Routing geometry	5	Four-chamber hourglass, Pinch event locus, Invariant eigen-axis, Swept phase arc, Trident 3-to-1 convergence...
Query architecture	5	state_at(t), next_event, events_in_cone, reachable relation query, identity reconstruction...
Numeric/fractal	4	Pascal parity recurrence, Sierpiński generation, Combination count C(5,3), Power-of-two scale boundary...
Glyph geometry	4	D-to-R cut and unroll, B-to-R lower-loop unfurl, Phi loop-plus-stem decomposition, Inward G/shelf transform...
Overlap geometry	4	Lens intersection region, Constructive interference interpretation, Tolerance/phase-slip buffer, Mutual-information domain...
Information architecture	4	Finite grammar, L-system production, Projective-phase state manifold, Typed state tuple...
Topology/control	3	One-bit parity narrow role, One-bit support mask, Deterministic jitter seed

Domain	Count	Representative mechanisms
Field geometry	3	Global hinge everywhere, Cone SDF, Sphere SDF
Metric geometry	3	Scaled 5–12–13 triangle, Leg ratio 2.4, Torque-cone angle
Persistence	3	External novelty log, Seed+grammar+log rebuild, Event lineage record
Architecture	3	Query-first, no mandatory frame loop, Optional projection, Schema-bound one-bit semantics
Coordinate geometry	2	Base-2 log-radius, Local spherical chart
Support geometry	2	Cone support, Nested shell support
Fluid threshold	2	Pythagorean-cup trigger, Greedy reset cycle
Topology geometry	2	Lemniscate crossing, Escher-style shrink to crossing
Identity	2	Lineage-based identity, Split/merge lineage
Pruning	2	Support rejection, Compatibility rejection
Numerics	2	Certified/bracketed root solve, Tangency and degeneracy handling
Indexing	2	Log-polar LUT, BST-to-flat-table translation
Risk	2	Algebraic closure, Event/branch explosion
Complexity	1	Horizon-independent closed expression
Memory	1	Novelty-proportional retention
Networking	1	Deterministic rollback/replay
Governance	1	Query authority and policy

5.1 Numeric, encoding and fractal layer

- Radix thresholds and positional weights.
- Explicit parser modes for literal binary glyphs.
- Sparse pulse placement with shifts and bitsets.
- Hamming weight and active-bit metadata.
- Pascal parity and Sierpinski generation through exact bit tests.
- Log-base-2 address displacement and quantization contracts.

5.2 Geometry and field layer

- D/B/phi cut-and-unroll morphology.
- Torque-axis deformation and split-oval waves.
- Overlap lenses as intersection, tolerance or interference regions.
- Cartesian, polar, log-polar and local spherical charts.
- SDF primitives, CSG, gradients, zero surfaces and guard bands.
- Analytic support cones, spheres, shells and local domains.

5.3 Topology and event layer

- Parity-controlled template/routing selection.
- Phase sheets and co-located non-coupling sectors.
- Möbius/Klein quotient maps with explicit orientation and ports.
- Hourglass partitions and central rank/event loci.
- Earliest valid event solving under support and compatibility.
- Transition, route, lineage and invariant updates.

5.4 GPU, memory and physical layer

- Typed hot-state ABI and packed records.
- SPIR-V compute modules and direct Vulkan pipelines.
- Dense diagnostic versus compact novelty paths.

- Device timestamps, cold/cached pipeline creation and validation.
- Fixed-point support/compatibility/guard datapath.
- Measured optofluidic B.C.E. endpoint and full-system accounting.

6. GPU-native compiler and runtime

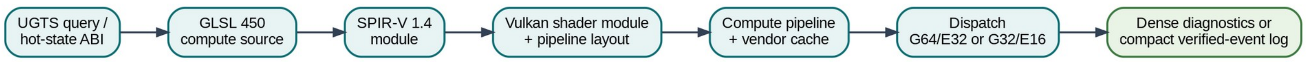


Figure 3. Native compilation and execution. The authoritative runtime is Vulkan/SPIR-V, not an engine adapter.

6.1 Compiled module set

Module	Bytes	SPIR-V	Local size	Entry
ugts_g32_evaluate.spv	5528	1.4	256x1x1	main
ugts_g32_evaluate_commit.spv	6148	1.4	256x1x1	main
ugts_g64_evaluate.spv	4888	1.4	256x1x1	main
ugts_g64_evaluate_commit.spv	5508	1.4	256x1x1	main

The runtime descriptor layout reserves set 0 bindings 0-2 for state, events and counters. Evaluate-only modules may optimize the unused counter binding out of the SPIR-V interface; evaluate+commit modules expose all three. The direct runtime creates shader modules, descriptor layouts, pipeline layouts, compute pipelines, buffers, command buffers, timestamp queries and pipeline caches without Unity or Godot.

REFERENCE QUERY CONVENTION

The bundled benchmark modules use target sheet 1, target orientation 0 and compatibility-mask bit 2. The exchange schema and CPU oracle permit explicit values. Production SPIR-V should expose query values through specialization constants, push constants or a query buffer, or compile a versioned fixed-query variant.

6.2 Evaluate versus evaluate+commit

Mode	Operation	Expected scaling behavior
evaluate	Each invocation writes its own event record.	No global atomics; maximum diagnostic throughput.
evaluate+commit	Also atomically increments candidate, support, compatibility and verified counters.	Contention grows with batch size and survivor density.
production compact	Ballot/prefix/append only verified events.	Reduces output memory; requires a measured compaction kernel or hardware FIFO.

6.3 Pipeline compilation and cache

Program	SPIR-V B	Module ms	Cold ms	Cache-seeded ms	Cache B	Reload
G64_E32_evaluate	4,888	0.0062	12.891	7.504	32	true
G64_E32_evaluate_commit	5,508	0.0029	9.544	7.279	32	true
G32_E16_evaluate	5,528	0.0034	8.548	14.652	32	true
G32_E16_evaluate_commit	6,148	0.0036	9.540	8.490	32	true

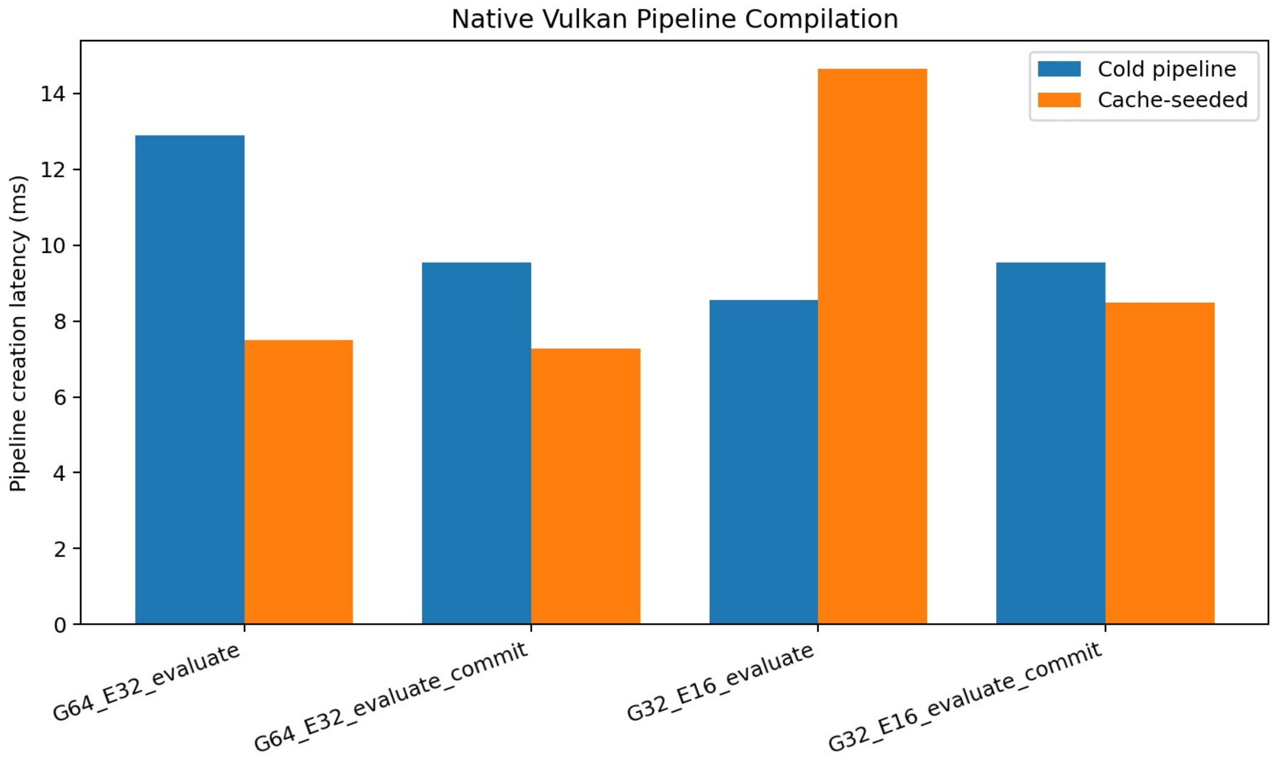


Figure 4. Pipeline-creation measurements on the named validation device. Cache-seeded creation is not guaranteed to be faster on every run or backend.

The 32-byte pipeline cache blobs are implementation-specific. They demonstrate cache extraction/reload plumbing, not a portable precompiled program format. The portable program object is SPIR-V plus the declared ABI.

7. State and event ABI

7.1 Authoritative G64/E32 profile

Record	Bytes	Contents	Use
G64State	64	12 float32 scalars plus four uint32 topology/lineage words.	Authoritative geometric reference and precision oracle.
E32Event	32	SDF, guard, confidence, time; verified, route, lineage hash, flags.	Dense diagnostic output and validation.

Offset	Bytes	Field	Meaning
0	16	position_time	x, y, z, query time
16	16	axis_radius	support axis x/y/z and radial reach
32	16	phase_guard	cone cosine, phase, epsilon, confidence floor
48	16	meta	sheet, orientation, compatibility mask, lineage seed

7.2 Packed G32/E16 profile

G32 stores twelve scalar values as IEEE binary16 pairs and packs topology into one word. E16 packs guard and confidence as binary16. This exactly halves dense state-plus-event traffic from 96 to 48 bytes per candidate, but only under a declared precision and guard-margin contract.

PRECISION RULE

Packed or fixed-point records are accepted only when coordinate, axis, radius and guard error remain below the declared event margin. A memory reduction is not a correctness proof.

7.3 Lineage

The 32-bit lineage hash in E32/E16 is a compact checksum for validation and routing. It is not a cryptographic identity and cannot replace the full generative address, branch history and ordered novelty log required for durable persistence.

7.4 Exchange schema

The package includes `spec/schema.json`, a JSON Schema for portable query/state/event exchange. It distinguishes the G64/E32, G32/E16 and Q16.16-squared hardware profiles and records the selected glyph-encoder variant.

8. Measured performance

MEASUREMENT CONTEXT

Runtime: direct Vulkan compute. Device: SwiftShader Device (Subzero). API: 1.3.0. Warmup: 3; measured iterations: 12; timing: device timestamps with 1.0 ns period. physical_gpu_claim=false.

Profile	Mode	p50 ms	p95 ms	p99 ms	CER M/s	SET M/s	ESB GB/s
G64_E32	evaluate	12.528	21.192	22.743	83.700	4.113	8.035
G64_E32	evaluate_commit	21.963	36.570	38.153	47.743	2.346	4.583
G32_E16	evaluate	13.307	18.860	21.321	78.798	3.748	3.782
G32_E16	evaluate_commit	19.172	32.585	33.743	54.693	2.602	2.625

8.1 Meaningful performance names

Symbol	Name	Value	Unit	Scope
CER	Candidate Evaluation Rate	83.6999	million candidates/s	G64_E32 evaluate, N=1048576
SET	Spherical Event Throughput	4.1125	million verified events/s	G64_E32 evaluate, N=1048576
ESB	Effective Substrate Bandwidth	8.0352	GB/s	G64_E32 evaluate, N=1048576
SRG	Support Rejection Gain	1.8982	x	G64_E32 deterministic corpus
CRG	Compatibility Rejection Gain	6.0019	x	G64_E32 deterministic corpus
EY	Event Yield	4.9134	%	G64_E32 deterministic corpus
SCR	State Compression Ratio	2.0000	x	N=1048576
ECR	Event Compaction Ratio	21.0228	x	G32_E16 evaluate, N=1048576
SNC	State-plus-Novelty Compression	2.9303	x	N=1048576
CCF	Commit Cost Factor	1.7531	x	G64_E32, N=1048576
PCP	Packed Compute Penalty	1.0622	x	N=1048576; software Vulkan device

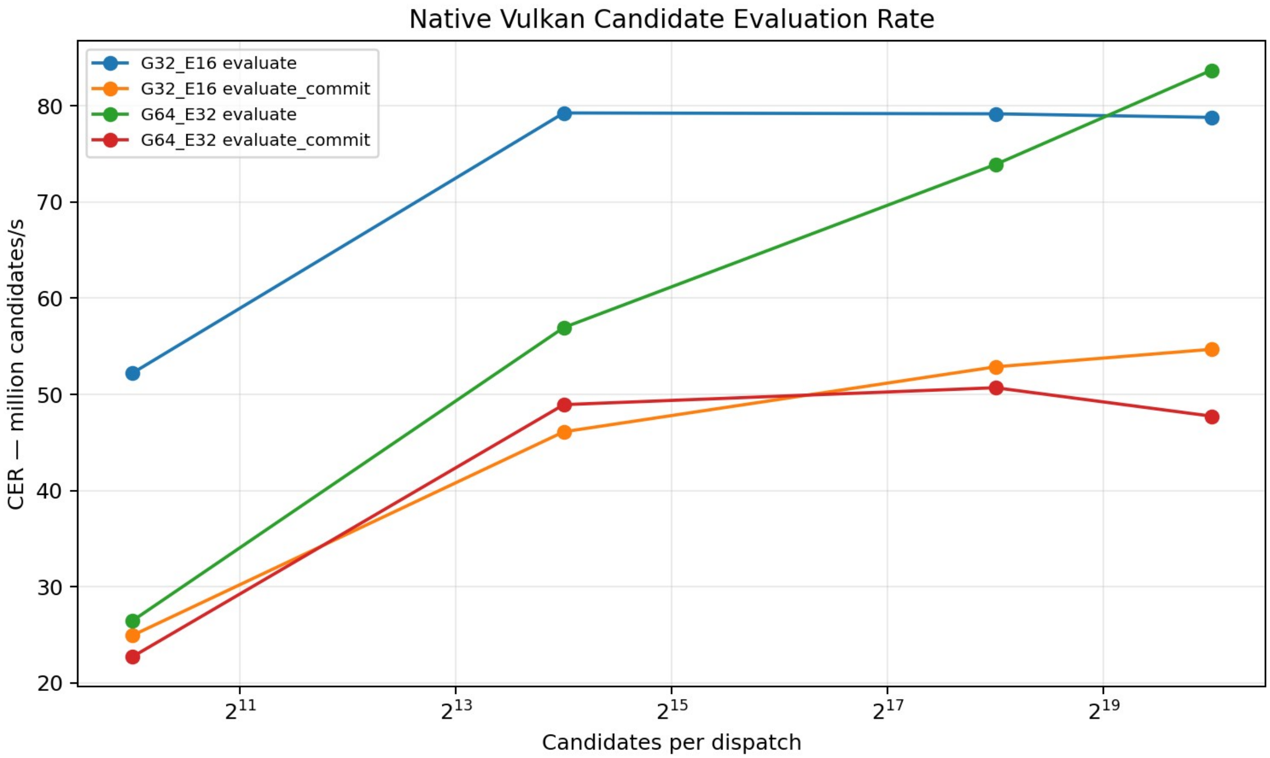


Figure 5. Candidate Evaluation Rate by profile, mode and batch size.

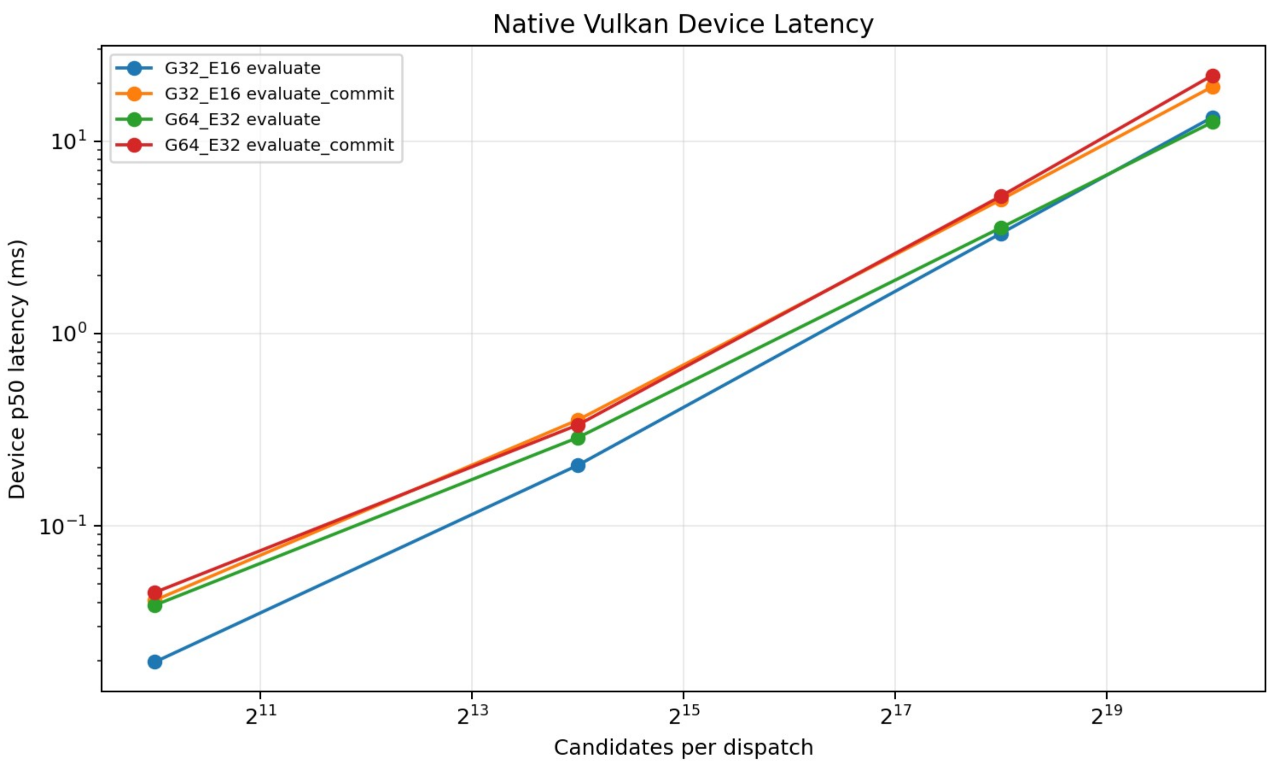


Figure 6. Device-timestamp dispatch latency. Tail latency is reported separately from p50.

8.2 Interpretation

At N=1,048,576, the G64/E32 evaluate path reached 83.700 million candidate evaluations/s and 4.113 million verified events/s on the software Vulkan device. Evaluate+commit cost 1.753x the evaluate p50 in G64/E32 because of global atomic counters.

The packed G32/E16 profile used half the dense memory but had a Packed Compute Penalty of 1.062x on this backend. Packing therefore reduced traffic without guaranteeing greater speed.

9. Memory and compression

Configuration	Bytes	Human	G64 dense	Retention
G64_E32 dense	100,663,296	96.000 MiB	100.000%	all states + dense outputs
G32_E16 dense	50,331,648	48.000 MiB	50.000%	all packed states + dense outputs
G32 state + compact E16 novelty log	34,352,480	32.761 MiB	34.126%	all packed states + verified events only
G32 compact E16 novelty log only	798,048	779.344 KiB	0.793%	verified events only; state must be rebuildable

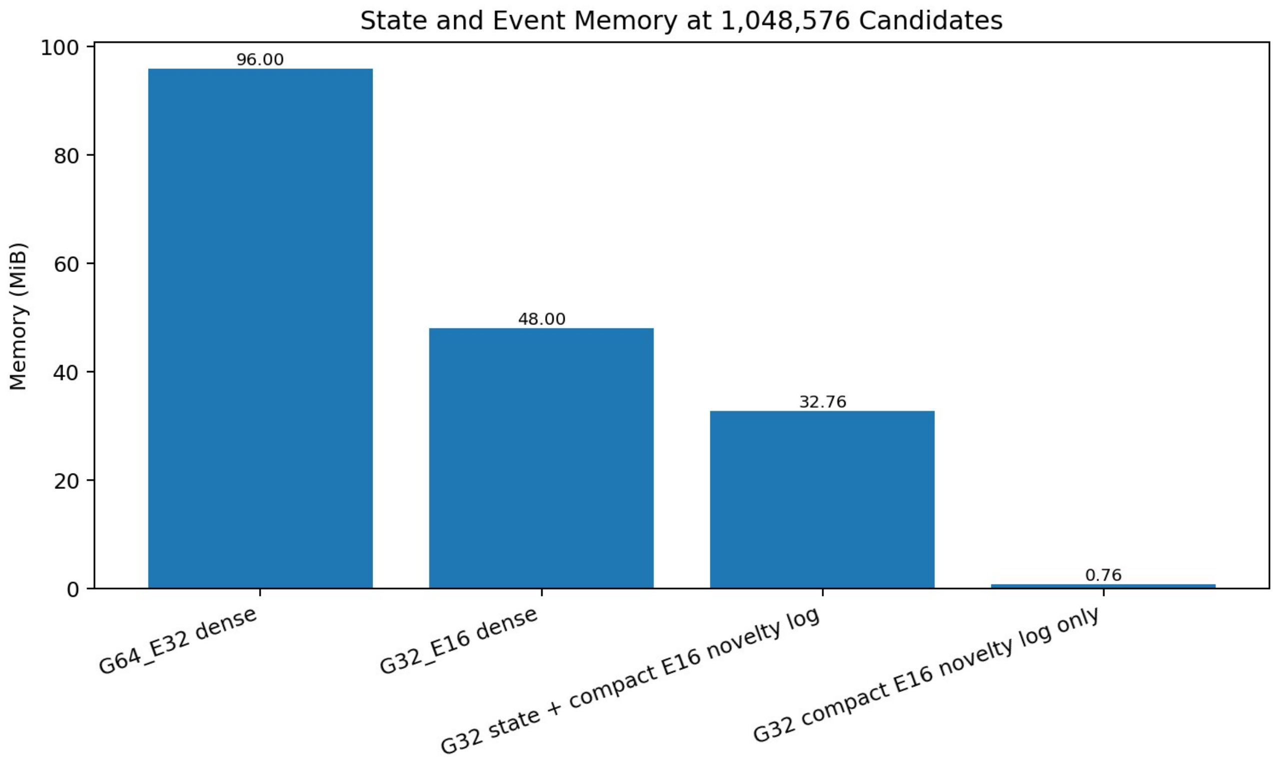


Figure 7. Memory footprint at 1,048,576 candidates.

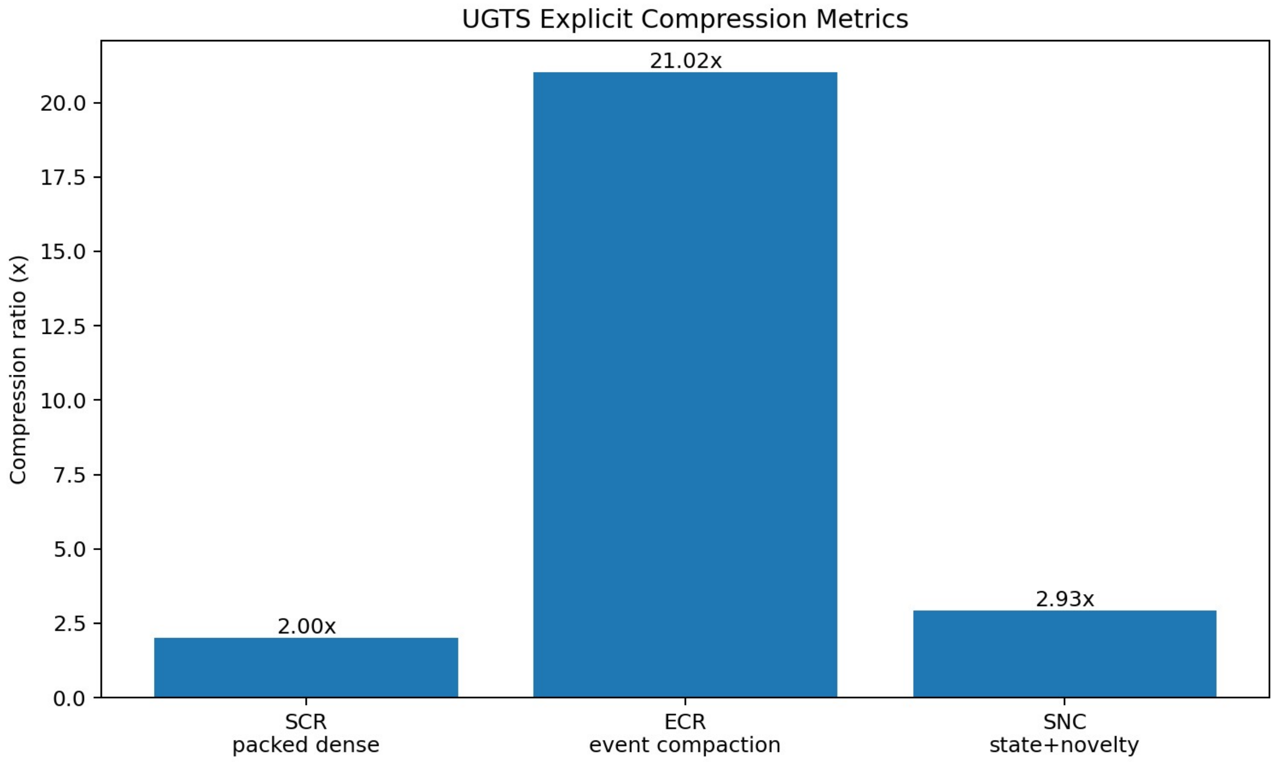


Figure 8. Named compression gains. ECR and SNC depend on observed event yield and reconstructibility.

9.1 Compression names and contracts

Metric	Formula	What it does not mean
State Compression Ratio (SCR)	G64/E32 dense bytes / G32/E16 dense bytes	Does not guarantee faster execution or equal numerical error.
Event Compaction Ratio (ECR)	dense output bytes / verified-only output bytes	Does not include compaction-kernel time; depends on event yield.
State-plus-Novelty Compression (SNC)	G64/E32 dense / (G32 state + compact E16 events)	Requires packed-state validity and verified-event compaction.
Novelty-only retention	verified events x E16 bytes	Valid only when state is reconstructible or authoritative state lives elsewhere.

9.2 One-bit masks

A one-bit support mask for 1,048,576 candidates occupies 131,072 bytes (128 KiB). It can cache a predicate result, but it cannot replace continuous state, topology, confidence or lineage.

10. Physical device architecture

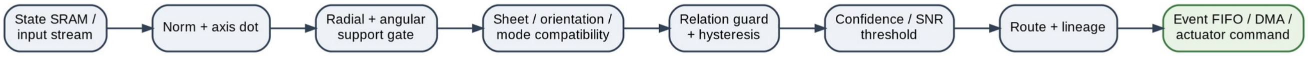


Figure 9. Fixed-function support-compatibility-guard-commit datapath for FPGA/ASIC or custom accelerators.

10.1 Fixed-function digital endpoint

The hardware mapping accepts a state stream or SRAM-resident records, evaluates radial and angular support, checks topology/mode compatibility, tests a relation guard, applies confidence/SNR, updates route and lineage, then writes a verified-event FIFO or DMA stream. It can answer event queries without a display or game engine.

10.2 Q16.16 squared gate

The bundled SystemVerilog reference avoids square roots. With a normalized axis, radial support uses r^2 <= R^2 and angular support uses dot>=0 plus dot^2 >= cos^2(alpha)*r^2. The guard is specified in squared-distance units. For a radial band epsilon, a useful conversion is guard_sq = 2*R*epsilon + epsilon^2 before fixed-point quantization.

10.3 Register and flow control

The portable register map exposes enable/reset/commit, target topology, mode bit, guard and confidence configuration, DMA bases, candidate count, four survivor counters, FIFO level, errors and timestamps. Production devices must use valid/ready flow control and treat FIFO overflow as a measurable missed-event condition.

10.4 Optofluidic and waveguide endpoint

The source hardware line becomes a bounded optofluidic front end: a local radial-angular input is shaped by a liquid lens or overclad, coupled into a compatible guided mode, observed at a balanced detector and declared as a Bounded Compatibility Event only when a measured guard crosses. [S05 pp. 2-10]

Physical layer	Authoritative model	Substrate responsibility
Optical field	Maxwell equations	Input support and declared measurement sector.
Liquid interface	Young-Laplace; fluid mechanics when transient	Actuator state, calibration and uncertainty.
Guided modes	Coupled-mode theory	Mode compatibility and transfer-state tags.
Detection	Detector/electronics noise and bandwidth	Guard, confidence, event timestamp and lineage.
Topology label	Explicit ports, orientation and transfer map	Routing abstraction only; no substitute for physics.

10.5 Hardware metrics

A physical device must report verified events/s, events/J, insertion loss, support and compatibility rejection, median/tail latency, false/missed events, timing jitter, drift, calibration interval, buffer memory and full-system energy. Propagation speed alone is not throughput.

11. Validation, corrections and kill criteria

11.1 Validation completed in this package

- Four SPIR-V modules parsed with the declared entry point, workgroup and descriptor decorations.
- Direct Vulkan shader-module and compute-pipeline creation completed.
- Pipeline-cache extraction and reload completed for all four programs.
- Device-timestamped dispatch completed at four batch sizes for four profile/mode combinations.
- All counter totals matched the CPU oracle.
- The first 4096 event records matched the CPU oracle for every benchmark case.
- 18 dependency-free reference tests passed.
- Device type was detected as CPU/software and physical_gpu_claim was recorded as false.

11.2 Corrections that preserve the useful substrate

Source claim/motif	Technical correction
100% equals 36	Standard value is 1; 36 belongs to a versioned custom glyph encoder.
1.5 as a fractional bit shift	Use multiplication or log2 displacement 0.5849625007.
0.680 bits -> 202.74 J	Dimensionally unsupported; energy requires a declared physical process and temperature.
Absolute vacuum cup/pinion	Pythagorean siphon normalized as threshold/reset with loss and hysteresis.
Klein bottle self-assembly	Retain only explicit quotient/gluing and routing abstraction.

Source claim/motif	Technical correction
One bit is complete state	One bit selects route/validity; continuous state, uncertainty and lineage remain.
Zero latency/heat/memory	Include compiler, cache, state, event log, source, actuator, detector, control and calibration.
Universal O(1)	Only fixed closed-expression state queries may be horizon-independent; event and branch costs remain.

11.3 Kill criteria

- Compatibility pruning fails to remove enough candidate relations.
- Event or branch density grows faster than avoided materialization.
- FP16/fixed-point error exceeds the guard margin or changes event ordering.
- Global atomic commit, compaction or FIFO backpressure dominates required throughput.
- Calibration energy/time or drift erases the physical advantage.
- False accepts, false rejects, misses or unordered events exceed the application budget.
- The digital control sidecar performs nearly all useful work.
- A conventional electronic reference is cheaper, faster or more accurate at equal error.

12. Implementation and reproduction

12.1 Package layout

Path	Purpose
spec/	197-mechanism catalog, schema, ABI layout, metrics and claims ledger.
gpu/	Shaders, SPIR-V, direct Vulkan runtime, bootstrap compiler, caches and scripts.
reference/	Dependency-free CPU oracle and binary packers.
tests/	18 executable unit tests.
benchmarks/	Raw JSON/CSV, derived metrics, plots and summaries.
hardware/	Device architecture, register map, acceptance plan and RTL reference.
sources/	Privacy-safe source register and extraction notes.
examples/	Query/event examples and custom encoder variants.
report/	This report and source document.

12.2 Reproduction commands

```
python -m unittest discover -s tests -v
bash gpu/scripts/run_all_validation.sh
```

On a physical Vulkan machine, unset `VK_ICD_FILENAMES` or point it at the vendor ICD, then rerun the direct benchmark. Record the selected device, driver, power mode and thermal state. Do not compare rates across devices without preserving profile, mode, batch size, precision and error budget.

12.3 Acceptance sequence

Steps 1-7	Steps 2-8
1. Select encoder and numeric profile.	2. Validate state ranges, axis normalization and guard margins.
3. Compile and hash the SPIR-V modules.	4. Create the Vulkan pipeline and persist the cache.
5. Run deterministic oracle vectors.	6. Benchmark p50/p95/p99 and named rates.
7. Measure memory, compaction and full energy.	8. Apply kill criteria before promoting a claim.

Appendix A. Complete 197-mechanism catalog

Each row is an atomic extracted or normalized mechanism. Source references use the privacy-safe S01-S10 register. Disposition and validation state prevent exploratory material from being confused with implemented or established operators.

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M001	Numeric/encoding	Radix digit threshold	S09 pp.1–3	A digit-count transition occurs at b^k in radix b.	RETAIN	Exact
M002	Numeric/encoding	Positional power weighting	S09 pp.1–4	Value is Σ digit_k·b^k.	RETAIN	Exact
M003	Numeric/encoding	Literal binary glyph parse	S01 pp.2–4; S09 pp.3–4	Parser mode is part of the schema; `$100_2=4_{10}$`.	RETAIN	Exact
M004	Numeric/encoding	Left-shift pulse placement	S01 pp.4–9	A pulse at bit position k has value 2^k.	RETAIN	Exact
M005	Numeric/encoding	Zero-based offset/ordinal distinction	S02 pp.1–4	Offset is zero-based; ordinal is offset+1.	RETAIN	Exact
M006	Numeric/encoding	Active-bit filtering	S09 pp.4–6	Extract set-bit positions or popcount; never change numeric identity silently.	RETAIN	Exact
M007	Numeric/encoding	Zero as inactive/silence metadata	S09 pp.4–5	A domain-specific interpretation attached to bit state, not arithmetic.	TRANSLATE	Schema-dependent
M008	Numeric/encoding	Binary 19 decomposition	S09 pp.3–5	19 = 16+2+1; Hamming weight 3.	RETAIN	Exact
M009	Numeric/encoding	Phonetic-to-active-count map	S09 pp.4–7	Associate an external label count with popcount under a declared language/tokenizer.	BOUNDED	Corpus-specific
M010	Numeric/fractal	Pascal parity recurrence	S09 pp.6–9	Cell parity is C(n,k) mod 2.	RETAIN	Exact
M011	Numeric/fractal	Sierpiński generation	S09 pp.6–14	Binary parity recurrence yields self-similar triangular occupancy.	RETAIN	Exact
M012	Numeric/fractal	Combination count C(5,3)	S09 p.6	Combinatorial count is 10; it is separate from value 19.	RETAIN	Exact
M013	Numeric/fractal	Power-of-two scale boundary	S09 pp.1–9	Dyadic levels define exact multiresolution boundaries.	RETAIN	Exact
M014	Glyph encoder	Standard percent semantics boundary	S01 pp.1,7–8	Default mathematical parser returns 1.0.	RETAIN	Exact
M015	Glyph encoder	Percent loop decomposition	S01 pp.2–4	Represent glyph as two loop primitives and one slash/stem primitive.	TRANSLATE	Constructive
M016	Glyph encoder	Slash/phi removal operator	S01 pp.2–4	Delete the slash primitive under an explicit glyph rewrite rule.	TRANSLATE	Rule-defined
M017	Glyph encoder	Open-loop terminal suppression	S01 pp.6–9	A source-specific encoder suppresses the final active bit when the terminal loop is open.	BOUNDED	Custom rule
M018	Glyph encoder	100100 concatenation	S01 pp.6–9	Concatenate two 3-bit pulse words to a 6-bit word.	RETAIN	Exact under custom schema
M019	Glyph encoder	Two-pulse timeline	S01 pp.9–10	Treat set-bit clusters as two events separated by an address interval.	TRANSLATE	Deterministic
M020	Glyph encoder	Pulse superposition by OR	S01 pp.6–10	Combine non-overlapping pulse masks with bitwise OR.	RETAIN	Exact
M021	Numeric/encoding	Explicit-width complement	S01 pp.18–22	Complement must specify width and mask; topology route is not unbounded `~x`.	TRANSLATE	Exact when width declared
M022	Numeric/encoding	Fractional scale correction	S01 pp.11–16	Multiplication by 1.5 corresponds to a log2 displacement of log2(1.5), not an integer bit shift.	CORRECTED	Exact
M023	Numeric/encoding	Packed fixed-width record	S06 pp.27–34; S07 pp.26–34	State fields have declared widths, scales and schemas.	TRANSLATE	Implemented
M024	Numeric/encoding	Fixed-point financial boundary	S06 pp.27–30	Use fixed-point/decimal for exact currency-like quantities; unrelated to geometry unless typed.	RETAIN	Established

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M025	Topology/control	One-bit parity narrow role	S05 pp.2,7; S03 pp.5–7	One bit selects a route, orientation or validity convention; continuous state remains separate.	RETAIN	Implemented
M026	Topology/control	One-bit support mask	S04 pp.1–2	A bit masks candidate sectors; it does not encode full geometry.	RETAIN	Implemented
M027	Topology/control	Deterministic jitter seed	S04 pp.7–10	Pseudo-random sequence is keyed by lineage/address for replay.	TRANSLATE	Specified
M028	Glyph geometry	D-to-R cut and unroll	S06/S07/S10 pp.1–2	A curve segment changes constraint class from loop arc to open segment while upper loop remains.	RETAIN	Constructive
M029	Glyph geometry	B-to-R lower-loop unfurl	S06/S07/S10 p.1	Release one loop edge and map it to an open leg.	RETAIN	Constructive
M030	Glyph geometry	Phi loop-plus-stem decomposition	S06/S07/S10 pp.1–2	Primitive graph with loop node and axial segment.	RETAIN	Constructive
M031	Kinematics	Stem as cutting/rotation axis	S06/S07/S10 pp.1–2	Axis defines local frame, cut plane and rotational reference.	RETAIN	Implemented
M032	Kinematics	Side-view pyramid as delta triangle	S06/S07/S10 p.2	Use a 2D triangular cross-section for a conical/pyramidal support.	TRANSLATE	Constructive
M033	Kinematics	Delta-T changing axis	S06/S07/S10 pp.2–4	Store reference axis and time derivative separately.	TRANSLATE	Specified
M034	Kinematics	Loop-to-line release	S06/S07/S10 pp.2–4	Topology flag selects closed relation vs open parametric segment.	RETAIN	Implemented as route
M035	Kinematics	Localized parity hinge	S06/S07/S10 pp.3–4	A graph edge toggles while a pivot node remains invariant.	RETAIN	Specified
M036	Field geometry	Global hinge everywhere	S06/S07/S10 pp.7–8	A field parameter switches or blends two implicit functions over the domain.	TRANSLATE	Specified
M037	Kinematics	Split zero-centered oval	S02 pp.10–12	Two half-curves share a zero-displacement boundary.	RETAIN	Constructive
M038	Kinematics	Antisymmetric torsional wave	S02 pp.12–13	Opposite signed displacement about the central hinge.	TRANSLATE	Physical model dependent
M039	Kinematics	2π unrolled into two π arches	S02 pp.13–15	Partition a periodic path into two half-period intervals.	RETAIN	Exact
M040	Kinematics	1:1 arch balance	S02 pp.13–15	Equal interval length; amplitude balance is a model parameter, not universal stability law.	BOUNDED	Conditional
M041	Overlap geometry	Lens intersection region	S02 pp.16–19	Explicit intersection `A∩B` with area/phase/confidence.	RETAIN	Exact geometry
M042	Overlap geometry	Constructive interference interpretation	S02 p.18	Combine complex amplitudes where supports intersect.	TRANSLATE	Model dependent
M043	Overlap geometry	Tolerance/phase-slip buffer	S02 p.18	Overlap width is a measurable alignment or phase-error margin.	TRANSLATE	Measurable
M044	Overlap geometry	Mutual-information domain	S02 pp.18–19	Use overlap membership as a join key; mutual information requires distributions, not area alone.	CORRECTED	Conditional
M045	Metric geometry	Scaled 5–12–13 triangle	S01 p.5	Multiply the Pythagorean triple by 0.3.	RETAIN	Exact
M046	Metric geometry	Leg ratio 2.4	S01 pp.5,15	Aspect/torque parameter `q=2.4`.	RETAIN	Exact
M047	Metric geometry	Torque-cone angle	S01 pp.15–16	Derive cone half-angle from leg ratio under stated triangle convention.	RETAIN	Exact under convention
M048	Kinematics	Logarithmic spiral step law	S06/S07/S10 pp.5–6	Scale changes exponentially with phase.	TRANSLATE	Parameter dependent
M049	Coordinate geometry	Base-2 log-radius	S01 pp.16–17; S04 pp.1–2	Multiplicative radial scale becomes additive coordinate displacement.	RETAIN	Implemented concept
M050	Coordinate geometry	Local spherical chart	S03 pp.5–7; S05 p.3	Use radial-angular coordinates locally around a query/coupler; not a global world replacement.	RETAIN	Canonical
M051	Support geometry	Cone support	S03 pp.5–7; S08 pp.15–18	Admit candidates satisfying radial reach and angular cosine threshold.	RETAIN	Implemented
M052	Support geometry	Nested shell support	S03 pp.3–6	Partition radius into ordered intervals for multiscale relevance.	RETAIN	Specified

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ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M053	Routing geometry	Four-chamber hourglass	S03 p.6; S08 pp.23–26	Four routing sectors meet at a central rank-changing locus.	TRANSLATE	Canonical abstraction
M054	Routing geometry	Pinch event locus	S03 p.6; S05 pp.2–5	A guard surface at which rank, route or mode changes.	RETAIN	Measurable
M055	Routing geometry	Invariant eigen-axis	S03 p.6; S08 pp.21–25	A tagged axis/mode remains invariant across route transition.	TRANSLATE	Specified
M056	Routing geometry	Swept phase arc	S03 p.6; S08 pp.23–25	Phase parameter maps between routing sectors.	TRANSLATE	Specified
M057	Routing geometry	Trident 3-to-1 convergence	S01 pp.25–27	Three input channels merge into one event/reset channel.	TRANSLATE	Constructive
M058	Fluid threshold	Pythagorean-cup trigger	S01 pp.27–29	Hysteretic threshold state machine with latch and reset.	RETAIN	Physical principle established; corpus mapping unvalidated
M059	Fluid threshold	Greedy reset cycle	S01 pp.27–29	On trigger, emit event then reset selected accumulator/state.	TRANSLATE	Specified
M060	Glyph geometry	Inward G/shelf transform	S06/S07/S10 pp.30–31	Additional glyph grammar rule; not core topology.	OPTIONAL	Exploratory
M061	Topology geometry	Lemniscate crossing	S09 pp.18–25	A crossing with orientation/chirality state and two lobes.	TRANSLATE	Exploratory
M062	Topology geometry	Escher-style shrink to crossing	S09 pp.20–27	Use bounded log-scale coordinate; do not infer infinite physical resolution.	CORRECTED	Bounded
M063	Field geometry	Cone SDF	S06/S07/S10 pp.7–8	Implicit conical support/boundary function.	RETAIN	Specified
M064	Field geometry	Sphere SDF	S03/S05; benchmark	$f(p)=\lVert p\rVert-R$ with zero guard.	RETAIN	Implemented and validated
M065	Topology	Möbius orientation reversal	S06/S07/S10 pp.4–6; S08 pp.10–14	A quotient boundary reverses orientation when traversed.	TRANSLATE	Mathematically exact abstraction
M066	Topology	Klein gluing map	S03 pp.2,6; S05 p.5; S08 pp.7–14	Specify ports and an orientation-reversing identification; use as routing, not literal container.	TRANSLATE	Canonical abstraction
M067	Topology	Same coordinate, different sheet	S03 p.6; S08 pp.15–19	State includes sheet/phase/address; coordinate equality alone does not imply coupling.	RETAIN	Implemented
M068	Topology	Non-orientable orientation bit	S03 pp.5–7	One bit records local orientation class and toggles at gluing maps.	RETAIN	Implemented
M069	Topology	Projected self-intersection	S06/S07/S10 pp.5–10	Treat apparent collision as two distinct sheet/address states unless physical interaction is specified.	CORRECTED	Specified
M070	Topology	Double vacuum as incompatibility	S03 p.6; S08 pp.6–18	No coupling because compatibility is false, despite co-location.	RETAIN	Canonical
M071	Topology	Phase-separated co-location	S03 p.6	Phase is part of state and coupling law.	RETAIN	Specified
M072	Topology	Four-way routing partition	S03 pp.6–7	Two route bits select one of four sectors around an event locus.	TRANSLATE	Specified
M073	Topology	Portal/gluing transition	S03 pp.5–7	At event, apply coordinate, sheet, orientation and lineage transform.	RETAIN	Specified
M074	Topology	Branch transition	S03 pp.5–8	Branch ID is updated only by a declared transition.	RETAIN	Specified
M075	Identity	Lineage-based identity	S03 pp.5–8,11–12	Identity is seed/generative address plus transition history/invariants.	RETAIN	Canonical
M076	Identity	Split/merge lineage	S03 pp.8,11–12	Splits create child addresses; merges require explicit reconciliation record.	RETAIN	Specified; stress test required
M077	Topology	Toroidal/poloidal chart labels	S01 pp.18–26	Use two periodic coordinates where appropriate; no claim of 4D physics follows.	TRANSLATE	Exploratory
M078	Topology	Re-entry map	S01 pp.18–28	Explicit transition maps output port to input address with optional orientation flip.	TRANSLATE	Specified
M079	Topology	Parity-controlled open/closed boundary	S06/S07/S10 pp.3–8	Parity selects one of two topology templates; changes occur at guarded events.	RETAIN	Specified

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M080	Topology	Sphere eversion analogy	S08 pp.10–12	Useful visualization only; not required by substrate.	DEMOTE	Analogy
M081	Topology	Physical Klein self-assembly	S06/S07/S10 pp.9–12	Not supported; requires material mechanics and cannot be inferred from SDF parity.	REJECT	Unsupported
M082	Topology	Topology-powered energy recycling	S01 pp.25–29	Topology does not create or preserve usable energy without a loss/drive model.	REJECT	Dimensionally unsupported
M083	Information architecture	Finite grammar	S03 pp.5–7,12	A bounded set of productions generates typed expressions/state.	RETAIN	Canonical
M084	Information architecture	L-system production	S06/S07/S10 pp.5–6	Parametric grammar operator with bounded depth and normalized output.	TRANSLATE	Depth must be bounded
M085	Information architecture	Projective-phase state manifold	S03 pp.5–7	Typed product state, not a single coordinate array.	RETAIN	Implemented subset
M086	Information architecture	Typed state tuple	S03 pp.5–7	Every field has unit, width and semantic convention.	RETAIN	Implemented
M087	Event calculus	Relation surface	S03 pp.5–7	An event candidate is a root of an explicit scalar relation.	RETAIN	Implemented sphere relation
M088	Event calculus	Support predicate	S03 pp.5–7; S05 pp.3,6	Local domain selects relevant candidates before event solving.	RETAIN	Implemented
M089	Event calculus	Compatibility predicate	S03 pp.5–7; S05 p.7	Coupling requires phase/sheet/address/mode/policy compatibility.	RETAIN	Implemented subset
M090	Event calculus	Earliest valid event	S03 pp.5–7	Choose earliest certified root satisfying all gates.	RETAIN	Prototype subset
M091	Event calculus	Transition operator	S03 pp.5–7	Apply atomic state patch at event and preserve invariants.	RETAIN	Specified
M092	Persistence	External novelty log	S03 pp.5–8,11–12	Closed dynamics may be rebuilt; unpredictable external changes are appended.	RETAIN	Implemented format
M093	Query architecture	state_at(t)	S03 pp.6–8	Evaluate closed expression at requested time without replay when possible.	RETAIN	Restricted families only
M094	Query architecture	next_event	S03 pp.6–8	Support + compatibility + root solve + branch resolution.	RETAIN	Restricted families only
M095	Query architecture	events_in_cone	S03 pp.6–7,11–12	Return events whose supports intersect query support.	RETAIN	Specified
M096	Query architecture	reachable relation query	S03 pp.6–7	Graph/grammar reachability under compatibility.	RETAIN	Specified
M097	Query architecture	identity reconstruction	S03 pp.7–12	Reconstruct identity without relying on current coordinate.	RETAIN	Specified
M098	Pruning	Support rejection	S03 pp.7–8; S05 p.6	Local analytic filter precedes expensive relation work.	RETAIN	Measured by SRG
M099	Pruning	Compatibility rejection	S03 pp.6–8; S05 pp.6–7	Filter supported candidates by typed mode/policy mask.	RETAIN	Measured by CRG
M100	Event calculus	Finite guard band	S05 pp.5–7; benchmark	Use epsilon band around relation zero to tolerate finite precision/noise.	RETAIN	Implemented
M101	Event calculus	Confidence threshold	S05 pp.3,6–7	Event commits only if confidence/certification exceeds floor.	RETAIN	Implemented synthetic model
M102	Event calculus	Hysteresis/debounce	S05 p.10; S01 pp.27–29	Separate enter/exit thresholds and hold time.	RETAIN	Specified
M103	Numerics	Certified/bracketed root solve	S03 pp.8–9,11–12	Return bounded root interval when exact analytic solve is unstable.	RETAIN	Future implementation
M104	Numerics	Tangency and degeneracy handling	S03 pp.8–11; S05 p.10	Classify derivative/rank conditions; avoid false crossing.	RETAIN	Risk register
M105	Field algebra	SDF zero as event boundary	S03 pp.6–7; S06 pp.7–8	Zero is semantic transition surface, not a reason to raymarch indefinitely.	RETAIN	Canonical
M106	Field algebra	CSG union	S04 p.2	Implicit union operator.	RETAIN	Exact for SDF sign

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M107	Field algebra	CSG intersection	S04 p.2; S02 pp.18–19	Implicit intersection/operator for overlap lens.	RETAIN	Exact for sign
M108	Field algebra	CSG subtraction	S04 p.2	Implicit difference.	RETAIN	Exact for sign
M109	Field algebra	SDF gradient orientation	S04 pp.2,4,9	Gradient provides normal/orientation; finite difference or analytic derivative.	RETAIN	Specified
M110	Field algebra	Parity-conditioned field blend	S06/S07/S10 pp.7–8	Discrete state selects or interpolates implicit geometry.	TRANSLATE	Specified
M111	Indexing	Log-polar LUT	S01 pp.16–22; S04 pp.1–8	Index radial-angular parameters by additive log radius and angle.	RETAIN	Specified
M112	Indexing	BST-to-flat-table translation	S01 pp.15–22; S06/S07/S10 pp.5–6	Compile tree decisions into breadth-first arrays or direct bit tests; avoid pointer chasing on GPU.	CORRECTED	GPU-specific
M113	Complexity	Horizon-independent closed expression	S03 p.8	For fixed expression size, state_at cost need not scale with skipped steps.	BOUNDED	Restricted claim
M114	Memory	Novelty-proportional retention	S03 pp.8,11–12	Store seed/grammar/invariants and external events, not every frame, when reconstruction is valid.	BOUNDED	Depends on reconstructibility
M115	Persistence	Seed+grammar+log rebuild	S03 pp.7–12	Deterministic replay from versioned seed, grammar and event log.	RETAIN	Specified
M116	Architecture	Query-first, no mandatory frame loop	S03 pp.1,5–12	Authoritative runtime serves state/event queries; projection is optional.	RETAIN	Canonical
M117	Architecture	Optional projection	S03 pp.1,9–10; S04	Render/print/display is a consumer of query results.	RETAIN	Canonical
M118	Event calculus	Bounded Compatibility Event	S05 pp.3,5–7	Verified event requires support, compatibility, guard crossing and confidence.	RETAIN	Implemented synthetic / specified physical
M119	Persistence	Event lineage record	S03 pp.5–8; S05 p.7	Event carries route, state flags, time/confidence and lineage.	RETAIN	Implemented
M120	Architecture	Schema-bound one-bit semantics	S05 p.7	Every bitfield is versioned and named by schema.	RETAIN	Implemented
M121	Networking	Deterministic rollback/replay	S08 pp.27–31; S03 pp.7–12	Replay versioned deterministic transitions plus external event log.	TRANSLATE	Restricted
M122	Risk	Algebraic closure	S03 pp.8,11	Composition must remain in a normalized relation family or lose direct-query advantage.	RETAIN	Open risk
M123	Risk	Event/branch explosion	S03 pp.8,11	Bound candidate count, branch depth and event density.	RETAIN	Open risk
M124	Governance	Query authority and policy	S03 pp.6,11	Policy/provenance is part of compatibility, not hidden side logic.	RETAIN	Specified
M125	GPU-native	GLSL-to-SPIR-V compilation	Package gpu/shaders + gpu/spirv	Compile compute kernels to SPIR-V; bundled modules are SPIR-V 1.4.	RETAIN	Compiled and inspected
M126	GPU-native	Direct Vulkan compute runtime	Package gpu/src/ugts_vulkan_bench.cpp	Create shader module, compute pipeline, descriptors, buffers and dispatch directly through Vulkan.	RETAIN	Executed
M127	GPU-native	Descriptor ABI	Package shader manifest	Set 0 binding 0 state, 1 event, 2 counters/commit.	RETAIN	Validated
M128	GPU-native	256-thread workgroup	SPIR-V manifest	One invocation evaluates one candidate.	RETAIN	Compiled
M129	GPU-native	G64 authoritative state record	Package spec	Four 16-byte vectors: position/time, axis/radius, phase/guard, metadata.	RETAIN	Implemented
M130	GPU-native	E32 full event record	Package spec	Float-bit scalars plus topology/lineage flags.	RETAIN	Implemented
M131	GPU-native	G32 packed state record	Package spec	Pair FP16 fields and pack topology metadata.	RETAIN	Implemented and validated
M132	GPU-native	E16 packed event record	Package spec	SDF plus packed guard/confidence, flags and lineage hash.	RETAIN	Implemented and validated
M133	GPU-native	Evaluate mode	Benchmark	Write per-candidate result without global counters.	RETAIN	Benchmarked

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M134	GPU-native	Evaluate+commit mode	Benchmark	Add four atomic counters for candidates/support/compatible/verified.	RETAIN	Benchmarked
M135	GPU-native	Device timestamp measurement	Benchmark	Use Vulkan timestamp queries and host fence timing.	RETAIN	Benchmarked
M136	GPU-native	Pipeline cache artifact	Benchmark	Persist vendor/driver-specific cache blob; portability is not assumed.	RETAIN	Reload tested
M137	GPU-native	SPIR-V module manifest	Package gpu/spirv/spirv_manifest.json	Hash and inspect version, entry point, local size and descriptors.	RETAIN	Validated
M138	GPU-native	Lineage hash	Shader code	Compact deterministic lineage checksum; not a cryptographic identity.	RETAIN	Validated
M139	GPU-native	Dense output path	Benchmark	Simple validation layout; non-events still occupy output.	RETAIN	Benchmarked baseline
M140	GPU-native	Verified-event compaction	Spec/derived metrics	Use ballot/prefix scan or atomic append to retain verified events.	TRANSLATE	Metric modeled; compact kernel not benchmarked
M141	GPU-native	Structure-of-arrays option	Spec	Split position/support/meta streams when hardware favors field locality.	OPTIONAL	Not benchmarked
M142	GPU-native	Memory coalescing contract	Spec	Records are 16-byte aligned; adjacent invocations access adjacent records.	RETAIN	Implemented
M143	GPU-native	FP16 error contract	Spec/tests	G32 requires declared range/tolerance; event guard must exceed quantization uncertainty.	RETAIN	Sample validated
M144	GPU-native	Cold versus cache-seeded compilation	Benchmark	Report shader-module, cold pipeline and cache-seeded pipeline times separately.	RETAIN	Benchmarked
M145	GPU-native	CPU reference oracle	Benchmark/tests	Scalar implementation checks counters and first 4096 outputs.	RETAIN	Passed
M146	GPU-native	Deterministic benchmark corpus	Benchmark	Fixed formula yields reproducible support, compatibility and event counts.	RETAIN	Passed
M147	GPU-native	No Unity/Godot target layer	User requirement	Engine adapters are intentionally absent; ABI is Vulkan/SPIR-V and physical register/stream definitions.	RETAIN	Package audited
M148	Projection	Cartesian-to-log-polar mapping	S04 pp.1–2	Transform only in output/sensor stage where radial symmetry helps.	RETAIN	Optional
M149	Projection	One-bit sector activation	S04 pp.1–2	Skip inactive sectors before field evaluation.	RETAIN	Optional
M150	Projection	Phasor coverage accumulation	S04 pp.2–3	Deterministic complex amplitude/phase accumulation or multisample coverage.	CORRECTED	Not quantum simulation
M151	Projection	Subpixel gradient alignment	S04 pp.4–6	Use gradient/normal to orient filter or coverage.	TRANSLATE	Optional
M152	Projection	Pulse-density modulation	S04 pp.4–6	Encode scalar magnitude as pulse density over a declared window.	RETAIN	Established technique
M153	Projection	Anisotropic delta-sigma	S04 pp.5–6	Shape quantization error by radial/angular sampling density.	TRANSLATE	Requires measurement
M154	Projection	Chromatic log-radius offset	S04 pp.7–8	A multiplicative radial scale becomes an additive log-radius offset per channel.	RETAIN	Mathematically exact coordinate identity
M155	Projection	Wavelength-dependent phase	S04 pp.7–8	Model dispersion with per-wavelength phase parameters.	TRANSLATE	Physical model required
M156	Projection	Seeded stochastic blend	S04 pp.7–10	Use deterministic noise threshold keyed by coordinate/lineage.	RETAIN	Optional
M157	Projection	Log-polar blue-noise screening	S04 pp.8–10	Warp screening frequency to maintain intended density under coordinate expansion.	TRANSLATE	Output-specific
M158	Projection	SDF edge lock	S04 pp.9–10	Near zero surface, bias threshold using gradient/coverage.	TRANSLATE	Output-specific

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M159	Projection	Dot-gain compensation	S04 p.10	Calibrate binary threshold to measured process gain.	RETAIN	Established output practice
M160	Projection	CMYK angular separation	S04 pp.9–10	Assign independent angular offsets to reduce moiré.	RETAIN	Output-specific
M161	Physical hardware	Liquid overclad	S05 p.4	Liquid changes effective index/coupling over a solid guide.	RETAIN	Prototype class established
M162	Physical hardware	Liquid-core guide	S05 p.4	Higher interaction with harder loss/packaging constraints.	OPTIONAL	Component class established
M163	Physical hardware	Input liquid lens	S05 p.4	Deformable interface couples free-space angular support into guide.	RETAIN	Component class established
M164	Physical hardware	Solid photonic substrate	S05 pp.4,8	Provides conventional low-loss routing and couplers.	RETAIN	Established
M165	Physical hardware	2×2 interferometric kernel	S05 pp.4,9	Minimal mode/phase comparison primitive.	RETAIN	Prototype proposal
M166	Physical hardware	Balanced detection	S05 pp.4,9	Difference readout reduces common-mode intensity noise.	RETAIN	Established
M167	Physical hardware	Spherical-to-mode overlap	S05 p.5	Coupling efficiency is normalized field overlap integral.	RETAIN	Established equation
M168	Physical hardware	Coupled-mode evolution	S05 p.5	Track propagation constants, attenuation and coupling.	RETAIN	Established theory
M169	Physical hardware	Young–Laplace curvature	S05 p.5	Pressure-curvature law drives lens shape.	RETAIN	Established theory
M170	Physical hardware	Maxwell field boundary	S05 p.5	Optical behavior must be solved/calibrated from dielectric geometry.	RETAIN	Established
M171	Physical hardware	Actuation modes	S05 pp.4,8	Actuator has range, settling, hysteresis, power and cycle life.	RETAIN	Measure
M172	Physical hardware	Measured B.C.E. guard	S05 pp.3,5–7	Detector threshold plus mode/policy compatibility commits event.	RETAIN	Prototype criterion
M173	Physical hardware	Calibration and drift state	S05 pp.6–10	Calibration coefficients and uncertainty are first-class state.	RETAIN	Required
M174	Physical hardware	Events per joule	S05 p.6	Count verified events over full source+actuator+detector+control energy.	RETAIN	Required
M175	Physical hardware	Optical insertion loss	S05 pp.6,10	Report loss beside event throughput.	RETAIN	Required
M176	Physical hardware	Packaging failure controls	S05 pp.8,10	Bubble rate, evaporation, swelling, alignment and seal life are kill criteria.	RETAIN	Required
M177	Physical hardware	Digital commit sidecar	S05 pp.4,7,9	Use FPGA/ASIC/MCU for thresholds, uncertainty, lineage and safety; this is not a game-engine adapter.	RETAIN	Canonical
M178	Physical hardware	Fixed-function support engine	Derived architecture	ASIC/FPGA datapath evaluates radial and angular support.	TRANSLATE	Architecture specified
M179	Physical hardware	Compatibility mask engine	Derived architecture	Bitfield compares before guard evaluation.	TRANSLATE	Architecture specified
M180	Physical hardware	Event FIFO and lineage commit	Derived architecture	Compact verified events and maintain monotonic sequence/lineage.	TRANSLATE	Architecture specified
M181	Physical hardware	Fluidic threshold/reset endpoint	S01 pp.27–29	Optional physical hysteretic reset, not a computation or energy source.	BOUNDED	Requires lab validation
M182	Physical hardware	Explicit topology transfer map	S05 p.5	Non-orientable naming is permitted only with port, orientation and transfer matrix.	RETAIN	Required
M183	Physical hardware	Failure/kill criteria	S05 p.10	Reject device if loss, drift, false events, actuation speed, packaging or sidecar cost erase benefit.	RETAIN	Required
M184	Audit	100% does not ordinarily equal 36	S01 pp.7–8	36 is output of a custom glyph encoder only.	CORRECTED	Exact
M185	Audit	100101 versus 100100 inconsistency	S01 pp.3–9	Canonical addendum uses the later explicit open-loop suppression rule; both encodings are logged as variants.	CORRECTED	Source inconsistency resolved explicitly

ID	Domain	Mechanism	Source	Normalized technical definition	Disposition	Validation
M186	Audit	Information entropy is not automatically heat	S01 pp.17–24	Thermodynamic energy requires temperature and irreversible erasure model; use Landauer lower bound only as a bound.	REJECT	Dimensionally invalid source claim
M187	Audit	Absolute single vacuum claim	S01 pp.25–29	A siphon uses pressure difference and gravity; no absolute vacuum is created.	CORRECTED	Established physics boundary
M188	Audit	Zero-cost chromatic correction	S04 pp.7–8	Log-space offset is cheap, not free; report instructions/bandwidth/passes.	REJECT	Absolute claim
M189	Audit	Perfect alias elimination	S04 pp.4–6	PDM/dither redistributes error; quality is finite and measurable.	REJECT	Absolute claim
M190	Audit	Universal O(1)	S03 pp.7–8; S08 pp.15–35	Only fixed closed-expression state queries may be horizon-independent; event/branch costs remain.	BOUNDED	Canonical correction
M191	Audit	Zero memory/latency/heat	S05 pp.2,10; S08	All state, caches, logs, sources, actuators and controls count.	REJECT	Canonical rejection
M192	Audit	Perfect determinism/exact chaos	S03 pp.8–11; S08	Finite precision, degeneracies and external novelty require uncertainty and logs.	REJECT	Unsupported absolute
M193	Audit	One bit as complete state	S05 pp.2,7; S08	One bit is route/validity only; continuous state and lineage remain separate.	REJECT	Canonical correction
M194	Audit	No broad phase ever	S08 pp.28–35	Support/compatibility are themselves analytic broad-phase/pruning operators.	CORRECTED	Terminology correction
M195	Audit	General AI/world replacement	S03 pp.1,7–12; S05 p.10	No support in corpus; restrict to state/event substrate.	REJECT	Unsupported
M196	Audit	Biological one-bit equivalence	S06/S07/S10 pp.13–24	Use as analogy only; biological/quantum mechanisms are not established by the geometry.	DEMOTE	Unsupported analogy
M197	Audit	Physical-GPU benchmark boundary	Package benchmark	Results prove API-native execution and correctness on named device, not physical-GPU throughput.	RETAIN	Explicit boundary

Appendix B. Claims ledger

The claims ledger records where the source is exact, schema-bound, translated, corrected or rejected. It is part of the technical deliverable, not a disclaimer added after the fact.

ID	Claim	Disposition	Source	Technical note
C001	Standard 100% equals 1	SUPPORTED	S01 pp.1,7–8	Ordinary arithmetic and source correction.
C002	Custom `100%` glyph encoder yields 36	BOUNDED	S01 pp.6–9	True only under explicit custom rule producing 100100_2.
C003	100101_2 yields 37	SUPPORTED VARIANT	S01 pp.3–4	Earlier encoder variant before terminal-bit suppression.
C004	Zero-based index 35 is ordinal position 36	SUPPORTED	S02 pp.1–4	Address/ordinal distinction; sum remains 35.
C005	19 has binary Hamming weight 3	SUPPORTED	S09 pp.3–5	$19=10011_2$.
C006	Three active bits equal three Dutch syllables	METADATA MAP	S09 pp.4–7	Corpus-specific correspondence, not numeric equality.
C007	Pascal parity yields Sierpiński structure	SUPPORTED	S09 pp.6–14	Exact parity construction.
C008	D/B can be morphed to R by cutting/unrolling a lower loop	CONSTRUCTIVE	S06/S07/S10 pp.1–4	Valid geometric rewrite, not a unique theorem.
C009	Parity bit can select open versus closed topology template	SUPPORTED AS COMPUTATION	S03; S06/S07/S10 pp.3–8	Bit is a route/template selector.

ID	Claim	Disposition	Source	Technical note
C010	SDF parity alone makes a physical Klein bottle self-assemble	REJECTED	S06/S07/S10 pp.9–12	No material mechanics or conservation model.
C011	Klein bottle is a useful routing/gluing abstraction	SUPPORTED AS ABSTRACTION	S03 pp.5–7; S05 p.5	Requires explicit ports, orientation and transfer map.
C012	Double vacuum means two stronger emptinesses	REJECTED	S03 p.6; S08	Normalized as non-coupling sectors.
C013	Double vacuum means co-located incompatible sheets	SUPPORTED AS MODEL	S03 p.6	Typed state and compatibility predicate.
C014	Universal constant-time world	REJECTED	S03 pp.7–8; S08	Only restricted fixed-expression horizon independence is plausible.
C015	Zero-memory world	REJECTED	S03 pp.7–12; S05 p.10	Seeds, grammar, state, caches and novelty logs consume memory.
C016	Memory can be proportional to irreducible novelty	BOUNDED	S03 pp.7–12	Only where closed dynamics are reconstructible.
C017	One bit encodes complete world state	REJECTED	S05 pp.2,7	One bit has a narrow route/validity role.
C018	Log-polar scaling turns magnification into additive radius offset	SUPPORTED	S04 pp.7–8	Coordinate identity $\rho = \ln r$.
C019	That correction is zero cost	REJECTED	S04 pp.7–8	Still costs arithmetic, fetches and bandwidth.
C020	Feynman vectors are required for rasterization	REJECTED TERMINOLOGY	S04 pp.1–3	Translated to deterministic phasor/coverage integration.
C021	PDM/dither eliminates aliasing	REJECTED ABSOLUTE	S04 pp.4–10	It shapes/redistributes finite error.
C022	Liquid lenses and optofluidic guides are component classes	SUPPORTED	S05 pp.4–12	Component feasibility, not system validation.
C023	Corpus system-level hardware advantage is already proven	REJECTED	S05 pp.10–12	Requires measured full-system baseline.
C024	0.680 bits correspond to 202.74 J	REJECTED	S01 pp.17–24	Dimensionally invalid without physical process; source number unsupported.
C025	Landauer bound can translate erased bits to a minimum heat at temperature T	SUPPORTED EXTERNAL ENGINEERING	Engineering correction	A lower bound, not actual device energy.
C026	Fractional bit shift by 1.5 is a native integer shift	REJECTED	S01 pp.11–15	Use scale multiply or log2(1.5) displacement.
C027	Pythagorean cup creates an absolute vacuum	REJECTED	S01 pp.27–29	It is a siphon driven by pressure/gravity.
C028	Pythagorean cup supplies a threshold/reset mechanism	SUPPORTED TRANSLATION	S01 pp.27–29	Useful hysteretic state-machine analogy and physical endpoint.
C029	Bundled SPIR-V is GPU-native intermediate code	SUPPORTED	Package artifacts	Modules validated for magic, entry, local size and bindings.
C030	Bundled benchmark proves physical GPU speed	REJECTED	Package benchmark	Device is SwiftShader CPU; only API-native correctness/performance on that device.
C031	G32/E16 halves dense record memory versus G64/E32	SUPPORTED	Package benchmark/spec	48 vs 96 bytes per candidate.
C032	G32/E16 is necessarily faster	REJECTED	Package benchmark	Packed decode was slower on the validation device.
C033	Evaluate+commit has no cost	REJECTED	Package benchmark	Atomic commit approximately doubles p50 on largest G64 batch.
C034	Verified-event compaction can reduce output memory by observed yield	SUPPORTED AS DERIVED METRIC	Package derived metrics	Requires a compaction kernel/FIFO; current dense benchmark does not measure it.

Appendix C. Metric and register dictionaries

C.1 Native metric dictionary

symbol	name	formula	required_context
CER	Candidate Evaluation Rate	N/device_time	device,batch,profile,mode,precision
SET	Spherical Event Throughput	V/device_time	support,compatibility,guard,confidence,error budget
ESB	Effective Substrate Bandwidth	N*(input_bytes+output_bytes)/device_time	logical traffic only; not raw DRAM bandwidth
SRG	Support Rejection Gain	N/N_support	support definition
CRG	Compatibility Rejection Gain	N_support/N_compatible	compatibility schema
EY	Event Yield	V/N	deterministic corpus or workload
CPC	Cold Pipeline Compilation	cold pipeline creation latency	driver/device/module
CPH	Cache-Seeded Pipeline Hydration	cache-seeded pipeline creation latency	cache portability restricted to matching implementation
NPF	Native Program Footprint	SPIR-V bytes + implementation-specific cache/binary bytes	module and driver
SCR	State Compression Ratio	dense_G64E32/dense_G32E16	same candidate count
ECR	Event Compaction Ratio	dense_output/verified_only_output	event yield; compaction implementation separate
SNC	State-plus-Novelty Compression	dense_G64E32/(G32_state+compact_E16)	state retained plus verified events
CCF	Commit Cost Factor	p50_evaluate_commit/p50_evaluate	same profile and batch
PCP	Packed Compute Penalty	p50_G32/p50_G64	same mode, batch and device

C.2 Hardware register map

Offset	Register	Access	Meaning
0x0000	CONTROL	R/W	bit0 enable; bit1 reset; bit2 commit; bit3 clear_counters
0x0004	STATUS	R	bit0 ready; bit1 busy; bit2 event_fifo_nonempty; bit3 overflow; bit4 calibration_valid
0x0008	PROFILE	R/W	0 G64/E32; 1 G32/E16; 2 Q16.16 squared gate
0x000C	MODE_BIT	R/W	compatibility mode bit 0..15
0x0010	TARGET_TOPOLOGY	R/W	bits7:0 sheet; bit8 orientation
0x0014	GUARD_CONFIG	R/W	fixed-point guard or profile-specific epsilon
0x0018	CONFIDENCE_MIN	R/W	float/fixed threshold or SNR code
0x001C	BATCH_COUNT	R/W	number of candidates in dispatch/stream batch
0x0020	STATE_BASE_LO	R/W	state buffer/DMA base address low
0x0024	STATE_BASE_HI	R/W	state buffer/DMA base address high
0x0028	EVENT_BASE_LO	R/W	event buffer/FIFO DMA base low

Offset	Register	Access	Meaning
0x002C	EVENT_BASE_HI	R/W	event buffer/FIFO DMA base high
0x0030	COUNT_CANDIDATE	R	processed candidates
0x0034	COUNT_SUPPORTED	R	support survivors
0x0038	COUNT_COMPATIBLE	R	support+compatibility survivors
0x003C	COUNT_VERIFIED	R	verified events
0x0040	EVENT_FIFO_LEVEL	R	queued compact events
0x0044	ERROR_FLAGS	R/W1C	overflow, invalid profile, numeric saturation, calibration/drift faults
0x0048	TIMESTAMP_LO	R	device timestamp low
0x004C	TIMESTAMP_HI	R	device timestamp high

C.3 State layout

profile	record	offset_bytes	size_bytes	field	encoding	meaning
G64	state	0	16	position_time	4xf32	x,y,z,t
G64	state	16	16	axis_radius	4xf32	axis x,y,z, radial reach
G64	state	32	16	phase_guard	4xf32	cone cosine, phase, guard epsilon, confidence floor
G64	state	48	16	meta	4xu32	sheet, orientation, compatibility mask, lineage seed
G64	event	0	16	scalar_bits	4xu32-as-f32	sdf, guard, confidence, event time
G64	event	16	16	topology_bits	4xu32	verified, route, lineage hash, state flags
G32	state	0	24	packed_float_pairs	6xu32	twelve IEEE binary16 fields
G32	state	24	4	packed_meta	u32	sheet[7:0], orientation[8], compatibility mask[24:9]
G32	state	28	4	lineage_seed	u32	lineage seed
G32	event	0	4	sdf_bits	u32-as-f32	float32 SDF
G32	event	4	4	guard_confidence	u32	two IEEE binary16 values
G32	event	8	4	state_flags	u32	verified, route, sheet, support, compatibility
G32	event	12	4	lineage_hash	u32	compact checksum

Appendix D. Source basis and package verification

D.1 Source register

ID	Source title	Pages	Status	Role	Privacy treatment
S01	Glyph-to-bit, 100% and timeline addendum	29	PRIMARY ADDENDUM	New addendum: custom glyph encoder, pulse pair, log-polar/SDF register model, trident/cup threshold motif	Personal contextual details omitted from synthesis
S02	Zero-based timeline, split oval and overlap geometry	19	PRIMARY	Index/ordinal distinction; torsional split-oval; double arches; overlap lens as interference, tolerance and shared domain	Narrative and political material excluded as non-technical
S03	Chronological synthesis of the spherical substrate line	12	CANONICAL	Mature interpretive rule: queryable relations, support, compatibility, event solving, transition, lineage, novelty log	None
S04	Vector graphical art system	10	TRANSLATED	Log-polar addressing, SDF/CSG, phasor-like coverage, PDM/delta-sigma, chromatic offsets and prepress output	System name retained only as corpus label
S05	Practical waveguide liquid-substrate lensing	12	CANONICAL HARDWARE	Bounded optofluidic endpoint, measured B.C.E., benchmark vector, fabrication and kill criteria	None
S06	D–phi–torque extended dialogue (title redacted)	52	EXPLORATORY / DEDUPED	Large exploratory branch: glyph morphing, parity hinge, SDF cone, Möbius/Klein motifs, fixed-width records, AR/waveguide and speculative biology	Original title contains personal identifiers; excluded
S07	D–phi–torque extended variant	42	EXPLORATORY / DEDUPED	Near-duplicate branch with a later grounded technical specification and dream/AR analogies	Personal/contextual passages omitted
S08	Hollowland double-vacuum exploration	36	MOTIF SOURCE / CLAIMS AUDITED	Source of double-vacuum, Klein, hourglass, event horizon, UUID, GPU-native and compression rhetoric	None
S09	19, radix thresholds and fractal topology	30	PRIMARY MOTIF SOURCE	Radix thresholds, active-bit/phonetic map, Pascal parity, Sierpiński, cone/lemniscate/chirality and log-polar motifs	Personal contextual details omitted
S10	D–phi–torque compact variant	34	EXPLORATORY / DEDUPED	Shorter near-duplicate of the D–phi–torque family; useful for glyph, SDF, fixed-width and waveguide motifs	Personal identifiers omitted

D.2 Package checks performed

- Source PDFs hashed; raw text excluded from package.
- Knowledge catalog count checked at 197 and IDs checked M001-M197.
- Python oracle and packers executed through 18 tests.
- SPIR-V modules regenerated from the shader/compiler path and inspected.
- Direct Vulkan run completed and outputs validated.
- Benchmark summaries and charts regenerated from the latest native JSON.
- Report rendered to PDF and visually inspected before packaging.

D.3 Reproducibility boundary

Driver program binaries and pipeline caches are vendor/backend specific. SPIR-V, shader source, ABI, oracle vectors and metric definitions are the portable contract. Physical-device claims require rerunning the package on the named target and measuring full-system energy, error and calibration.

D.4 Canonical shorthand

```
local support -> compatibility -> guard crossing
-> verified event -> route/transition -> lineage + novelty log
```